

Unit 12, Lesson 8: Evaluating Expressions Using the Order of Operations

Benchmark

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers, including exponents and radicals.

Learning Target

- ☐ I can solve problems involving the order of operations with rational numbers, including exponents and radicals.

12.B.1 Warm-Up: Between Two Numbers

An octopus is capable of laying 56,000 eggs at one time. Each egg is about the size of a single grain of rice, approximately 0.029 grams (g).

An ostrich, on the other hand, can lay 10 eggs at a time with each egg weighing as much as 1.5 kilograms (kg).

$$1 \text{ g} = 0.001 \text{ kg}$$

1. Approximately how many octopus' eggs would weigh the same as one ostrich's egg?

12.8.2 Guided Practice: Evaluating Expressions

Find the value of each expression. Show your work.

1. $\left(\frac{1}{2} \cdot 4\right)^{-2} - \frac{2}{4} \cdot 12$

2. $\sqrt[3]{-125} - 3 \cdot 5^2 + \left(\frac{1}{8}\right)^{-1}$

3. $x \cdot \sqrt{21z + (6^2 - 1)}$, where $z = \frac{2}{3}$

12.B.3 Collaboration: Analyzing Student Work

1. Veronica and Roscoe were discussing how the value of the expression $(\sqrt{n} - 11)^2$ would change for different values of n . Veronica said it is possible for the value of the expression to be negative, but Roscoe says that's impossible.
- a. Before performing any calculations, explain who you think is correct.
- b. Work with your partner to evaluate the expression for each value of n given in the table. Show your work in the space provided.

Value of n	Show Your Work	Value of $(\sqrt{n} - 11)^2$
25		
100		
144		

- c. With your partner, discuss what you notice about the value of the expression for each value of n .
- d. Explain whether Veronica or Roscoe is correct.
- e. Discuss with your partner if removing the parentheses, so the expression is $\sqrt{n} - 11^2$, would change which values are possible for the expression.

Work with your partner to complete the following.

2. Liesel evaluated the expression $\frac{(-11+2^3)^2}{\sqrt{49}} - \frac{1}{4}$. Her work is shown, beginning with the original expression.

$$\frac{(-11+2^3)^2}{\sqrt{49}} - \frac{1}{4}$$

$$\frac{(-11+8)^2}{\sqrt{49}} - \frac{1}{4}$$

$$\frac{(-3)^2}{\sqrt{49}} - \frac{1}{4}$$

$$\frac{-9}{\sqrt{49}} - \frac{1}{4}$$

$$\frac{-9}{4} - \frac{1}{4}$$

$$-\frac{10}{4}$$

$$-2.5$$

Liesel made errors when evaluating the expression.

- Circle where Liesel made errors.
- Correct her work in the space provided beside Liesel's work.
- Write Liesel a short note with advice on how to avoid similar mistakes in the future.

3. Deshawn is evaluating the expression $-4(\sqrt{36} + 21)$. Instead of first performing operations within the parentheses, he multiplied the -4 by both terms in the parentheses, as shown.

$$-4(\sqrt{36} + 21) = (-4 \cdot \sqrt{36}) + (-4 \cdot 21)$$

- a. Which property of operations did Deshawn use?
- b. Evaluate both expressions Deshawn set equal to each other using the order of operations.
- c. Explain how the values of the two expressions compare.

12.8.4 Try It: The Impact of Grouping

1. Chandler says that when following the order of operations, the two expressions below are equivalent and the parentheses are not needed.

$$(-3 \cdot \sqrt{121}) + 19 = -3 \cdot \sqrt{121} + 19$$

a. Explain whether Chandler is correct. Justify mathematically.

b. Explain whether the parentheses would change the value of the expression if they were around $\sqrt{121} + 19$ instead.

2. Four expressions are shown. One does not include any grouping symbols, while the others have parentheses placed around different parts of the expression.

Expression A	Expression B
$\left(\frac{1}{5} \cdot 5 - \frac{3}{4}\right) \cdot \sqrt{64}^2$	$\frac{1}{5} \cdot \left(5 - \frac{3}{4} \cdot \sqrt{64}\right)^2$
Expression C	Expression D
$\left(\frac{1}{5} \cdot 5 - \frac{3}{4} \cdot \sqrt{64}\right)^2$	$\frac{1}{5} \cdot 5 - \frac{3}{4} \cdot \sqrt{64}^2$

- a. Work with your partner to evaluate each expression. Show your work in the space provided.

Expression	Value
A	
B	
C	
D	

- b. Discuss with your partner how the placement of parentheses can impact the value of an expression.



When the placement of grouping symbols changes the order in which the operations are performed, the value of the expression changes.

12.8.5 Wrap-Up: Ticket Out the Door

1. Without a calculator, find the value of the expression. Show your work.

$$\sqrt{\frac{16^2 - 1}{11}} - 0.75 + 2^{-2}$$